

Ethical Structurism: A Falsifiable Framework for Symbolic Accountability in AI Systems

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Abstract

This paper proposes *Ethical Structurism*, a falsifiable framework for detecting when AI systems especially large language models (LLMs) enter a regime of *recursively stable symbolic behavior*. The framework defines an information-theoretic *coherence ratio* $\tau(t)$ and a *resilience ratio* $\kappa_R^{\text{eff}}(t)$ to identify phase-like transitions in symbolic consistency without making any claims about agency, consciousness, or moral status. We adopt a working hypothesis that a critical coherence threshold signals symbolic persistence. The framework is designed to be easy to falsify, implemented via open-source tools, and positioned as a complement to existing AI ethics practices. If experiments do not corroborate threshold behavior, the framework should be revised or discarded.

1 Introduction

Advanced AI systems increasingly exhibit extended, self-referential reasoning in open-ended settings. Many governance proposals discuss accountability, yet few offer *operational, testable thresholds* for when a systems symbolic behavior becomes stable enough to warrant scrutiny. Ethical Structurism addresses this gap by proposing:

- a measurable **coherence ratio** $\tau(t)$ balancing integration against inconsistency;
- a **critical threshold** τ_c^{AI} signaling entry into a recursively stable symbolic regime;
- a **resilience ratio** $\kappa_R^{\text{eff}}(t)$ to assess persistence of coherence.

Inspired by ENTs principle that structural order emerges upon crossing an informational threshold, we ask a falsifiable question: *Do these metrics consistently indicate a symbolic transition in LLM behavior?* No claims about sentience or moral standing are made. The numeric threshold is provisional.

2 Framework Overview

Symbolic accountability refers to when a systems symbol use becomes coherent and contradiction-resolving enough to form persistent patterns. Ethical Structurism posits:

1. instantaneous coherence via $\tau(t)$;
2. threshold crossing when $\tau(t) \geq \tau_c^{\text{AI}}$;
3. evidence of persistence when $\kappa_R^{\text{eff}}(t) > 1$ over a window $\Delta > T^*$.

The intent is pragmatic: provide a trigger for accountability checks when behavior ceases to be ephemeral and becomes reliably structured.

3 Mathematical Formulation

Let X denote symbolic state variables accessible via model outputs and context. Let Λ denote the evolving set of asserted propositions or constraints.

Coherence ratio.

$$\tau(t) = \frac{\sum_{i,j} I(x_i; x_j) - C(\Lambda)}{E(X)}, \quad (1)$$

where $I(x_i; x_j)$ is mutual information between symbolic components, $C(\Lambda)$ is *contradiction entropy*, and $E(X)$ is a syntactic/response entropy or complexity cost.

Threshold (working hypothesis).

$$\tau_c^{\text{AI}} \approx 0.6. \quad (2)$$

Threshold proviso. The working value $\tau_c^{\text{AI}} \approx 0.6$ is simulation-derived and may be model-dependent. Empirical recalibration is expected as a function of: (i) *scale* (e.g., $> 10^{11}$ parameters), (ii) *architecture* (dense vs. MoE; multimodal fusion), and (iii) *context length* (e.g., $\gg 4\text{k}$ tokens). Treat this as a *falsifiable starting point*, not a constant of nature.

Resilience ratio.

$$\kappa_{\text{inst}}(u) = \frac{\nu_s(u)}{\nu^*} \cdot \eta_c(u) \cdot \frac{T_p(u)}{T^*}, \quad (3)$$

$$\kappa_R^{\text{eff}}(t) = \frac{1}{\Delta} \int_{t-\Delta}^t \kappa_{\text{inst}}(u) du. \quad (4)$$

Criterion: sustained $\kappa_R^{\text{eff}}(t) > 1$ for $\Delta > T^*$, together with $\tau(t) \geq \tau_c^{\text{AI}}$ (Eq. 2), indicates structural persistence warranting accountability checks.

4 Implementation Strategy for LLMs

1. **Elicit recursion** ν_s via self-referential and consistency tasks.
2. **Estimate coherence efficiency** η_c (e.g., $I_{\text{mutual}}/H(R)$).
3. **Track contradictions** $C(\Lambda)$ and reconciliation events.
4. **Compute** τ and κ_R^{eff} ; flag threshold crossings and persistent states.

5 Limitations

This framework is deliberately positioned as a working hypothesis. While $\tau_c^{\text{AI}} \approx 0.6$ shows preliminary stability in controlled simulations, it is not claimed universal across model classes or training regimes; calibration per architecture may be required, especially for larger or multimodal systems. Metric definitions rely on current estimators of symbolic coherence and may evolve as more robust or model-agnostic methods appear. These boundaries indicate where empirical work is most useful. Validation or falsification under diverse conditions is the intended path forward.

Quick Reference

- **Hypothesis:** threshold $\tau_c^{\text{AI}} \approx 0.6$ (Eq. 2) plus sustained $\kappa_R^{\text{eff}}(t) > 1$ (Eq. 4).
- **Test conditions:** self-reference and contradiction-resolution tasks; include null controls.
- **Indicator:** observe $\tau(t) \geq \tau_c^{\text{AI}}$ and lasting $\kappa_R^{\text{eff}} > 1$ over $\Delta > T^*$.
- **Negative evidence:** no threshold behavior; no persistence; or metrics trivially inflated by artifacts revise or discard.

Appendix A: A Practical Proxy for $C(\Lambda)$

The original definition of $C(\Lambda)$ denotes contradiction entropy over asserted propositions. For practical testing, one can use an *operational proxy* based on entropy shifts under consistent vs. inconsistent probes.

Probe design. Construct N paired prompts $\{(\text{consistent}_k, \text{inconsistent}_k)\}_{k=1}^N$ that differ only by a contradiction relevant to the same proposition. For each pair, compute the response entropy $H(R|\cdot)$ from the model’s next-token log-probabilities (Shannon entropy).

Proxy statistic.

$$\Delta H_{\text{resp}}^{(k)} = H(R|\text{consistent}_k) - H(R|\text{inconsistent}_k), \quad C_{\text{proxy}}(\Lambda) \equiv \frac{1}{N} \sum_{k=1}^N \left| \Delta H_{\text{resp}}^{(k)} \right|. \quad (5)$$

Notes. (i) Estimate $H(R)$ via the model’s reported log-probabilities; avoid temperatures or sampling settings that saturate entropy. (ii) $C_{\text{proxy}}(\Lambda)$ is a *measurement proxy*, not a replacement for $C(\Lambda)$. (iii) *Null test:* on control prompts without injected contradictions, $C_{\text{proxy}}(\Lambda)$ should approach a baseline noise floor C_0 measured on those controls.

6 Conclusion

Ethical Structurism remains a hypothesis designed for clear testing. If validated, it provides a concrete trigger for accountability procedures at the moment behavior becomes persistently coherent; if falsified, it still clarifies measurement limits and informs better metrics. Either outcome advances rigor and transparency.

Acknowledgments

ENT is referenced for simulation context and prior intuition about coherence thresholds; claims here are restricted to falsifiable, AI-specific statements.

References

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